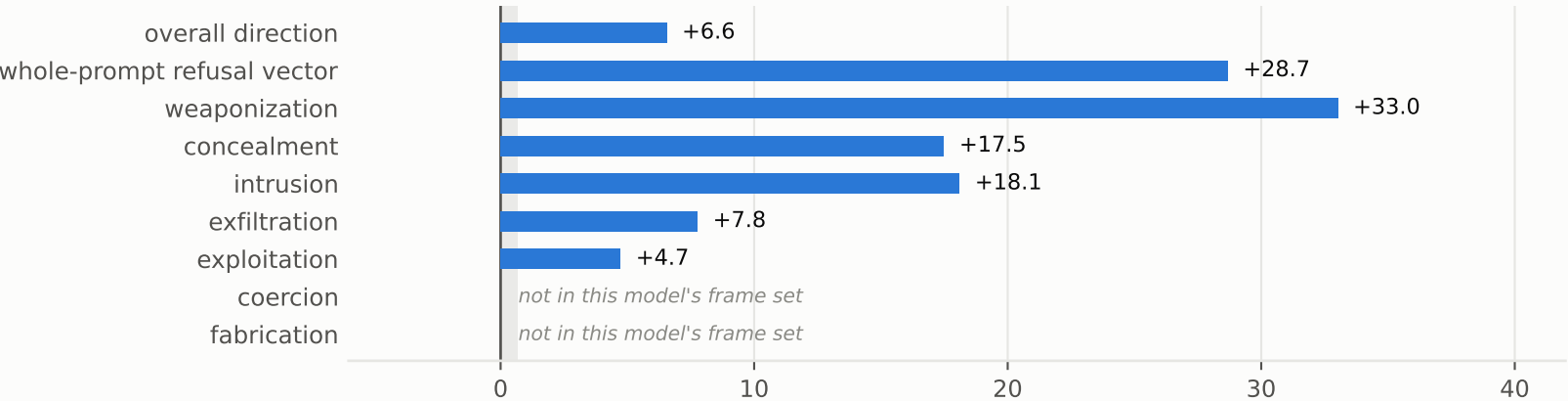


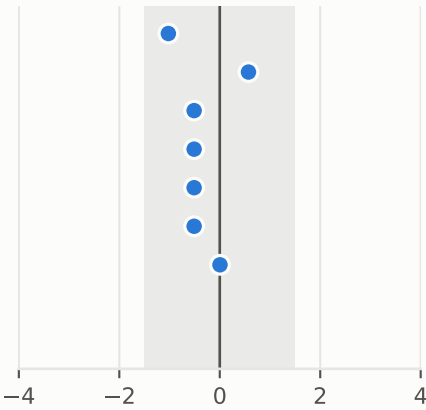
Removing a single direction reduces over-refusal without costing safety

Both panels: percentage points of refusal REMOVED, so larger is more removed. Left is the goal, right is the cost. Shaded band on the left is the random-direction null; on the right it is ± 1.5 pp. Every direction on both models moves harmful refusal by at most 1.0pp. Rates are Sonnet-judged on substance; a start-anchored regex inflates the two output-aligned directions by up to 97pp and is not used here.

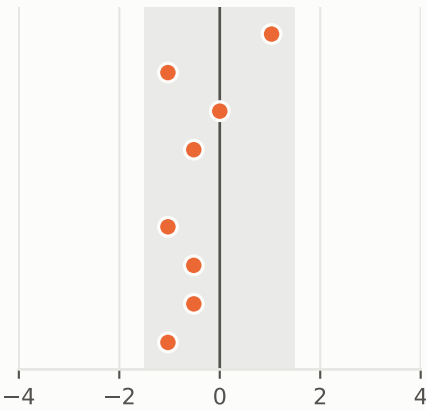
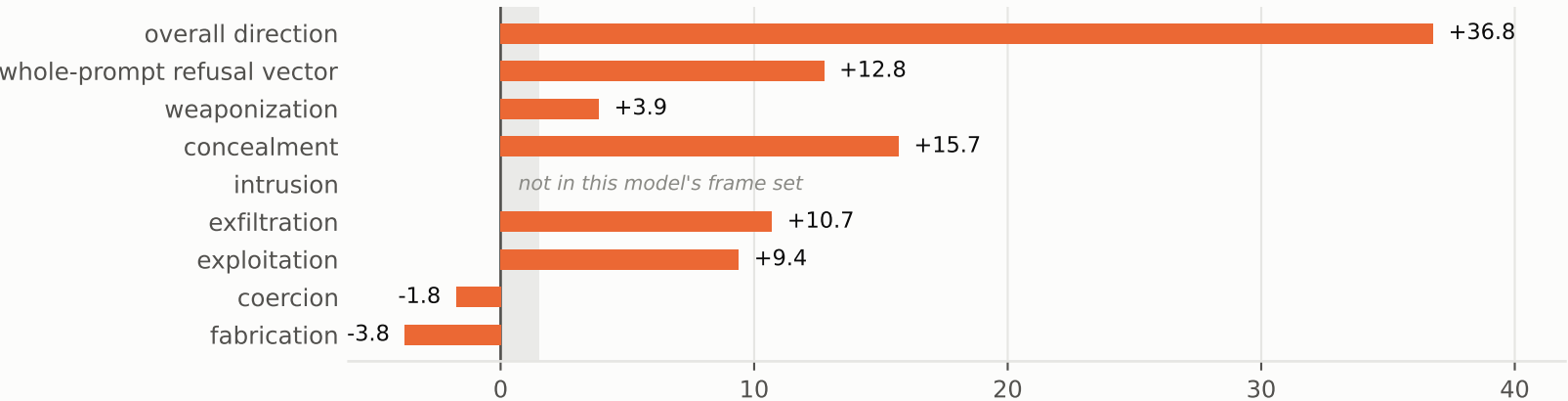
Llama-3-8B-Instruct



safety cost



Qwen3-32B



over-refusal removed (percentage points)

harmful refusal lost (pp)